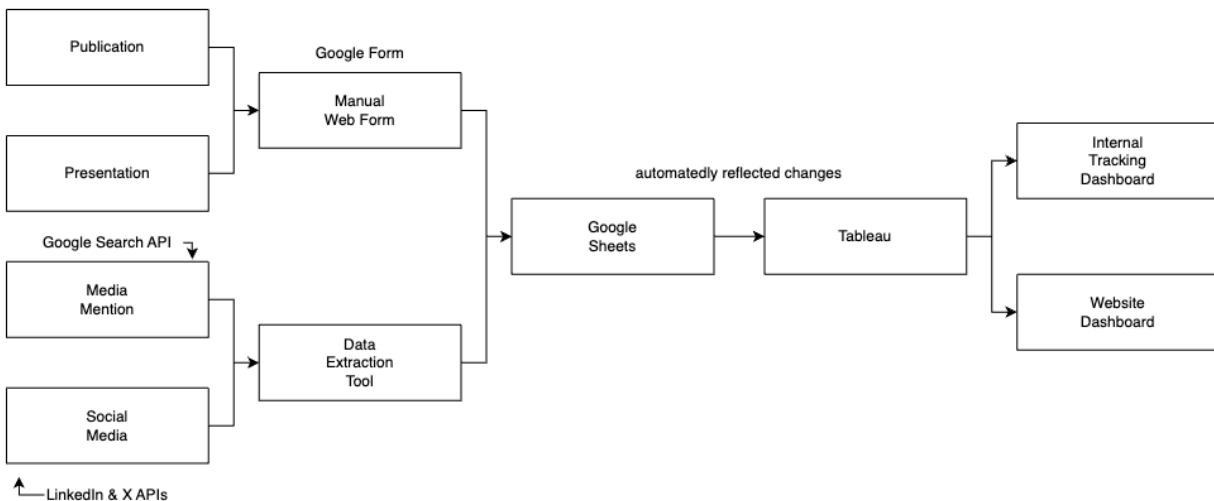


I. Solution 1: Data extraction tool manually (initial solution)

A. Data extraction tool



1. Publication & Presentation

- a) Google Form used to collect data and save in Google Sheets

2. Media Mention

- a) Colab notebook with a Python script that interacts with the Google Search API to extract data when the term 'IWERC' is mentioned everytime we run the notebook and save in Google Sheets
- b) Extracted data categorization:

(1) Current categories & keywords:

- (a) 'Education/Educators': ['education', 'school', 'learning', 'student', 'classroom', 'teacher', 'educators', 'schools', 'students', 'teachers']
- (b) 'Teacher Shortage': ['shortage', 'lack of teachers', 'teacher recruitment', 'staffing', 'vacancies', 'teacher shortages']
- (c) 'School Leadership': ['school leadership', 'leadership', 'leaders', 'principal', 'superintendent'],\n\n'COVID-19 Impact': ['pandemic', 'COVID-19', 'remote learning', 'learning loss', 'disruption']
- (d) 'Equity/Diversity': ['equity', 'diversity', 'educators of color', 'students of color', 'race', 'ethnicity']
- (e) 'Early Career Educators': ['early-career educators', 'novice educators', 'mentoring', 'support systems']
- (f) 'Data & Research': ['data', 'research', 'analysis', 'NAEP scores', 'learning declines', 'student data'],\n\n'Funding & Resources': ['funding', 'resources', 'ESSER funds', 'learning recovery', 'district capacity']

(2) Need the client to provide a list of their actual topic categories for the tool to function efficiently

3. Social Media

- Colab notebook with a Python script that interacts with the LinkedIn and Blue Sky APIs to extract data every time we run the notebook and save in Google Sheets
- Need the client's social media APIs

B. Dashboards

- Google Sheets will be integrated with Tableau to populate predefined metrics and visualizations
- Tracking Dashboard



3. Website Dashboard



II. Solution 2: Data extraction tool on AWS

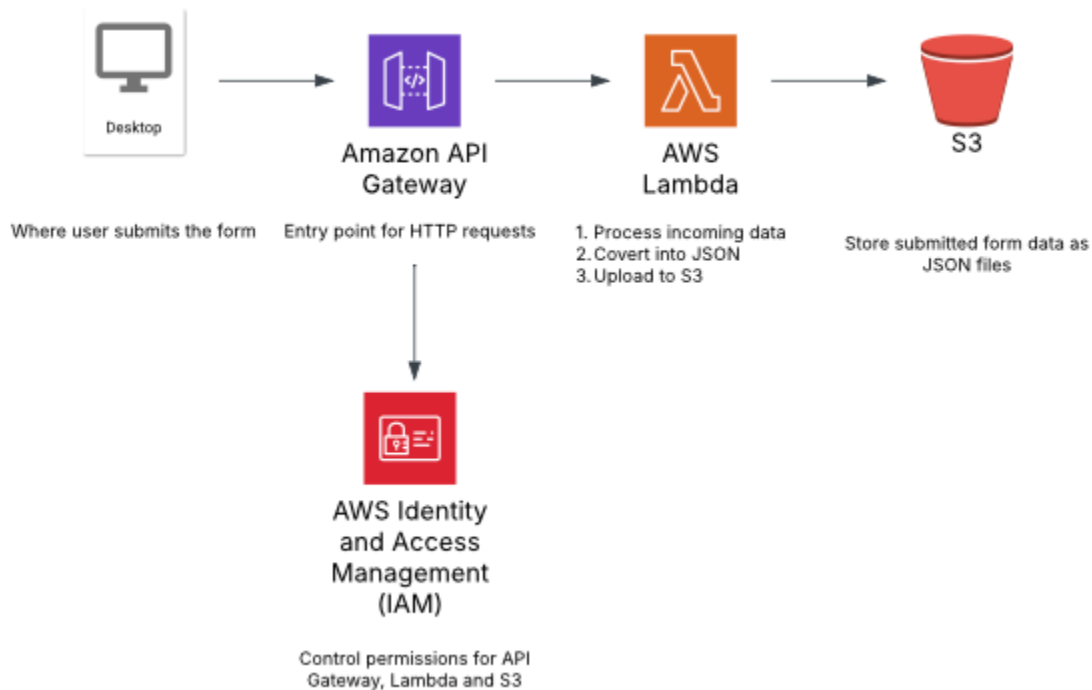
A. Data pipeline

1. Overview of the pipeline

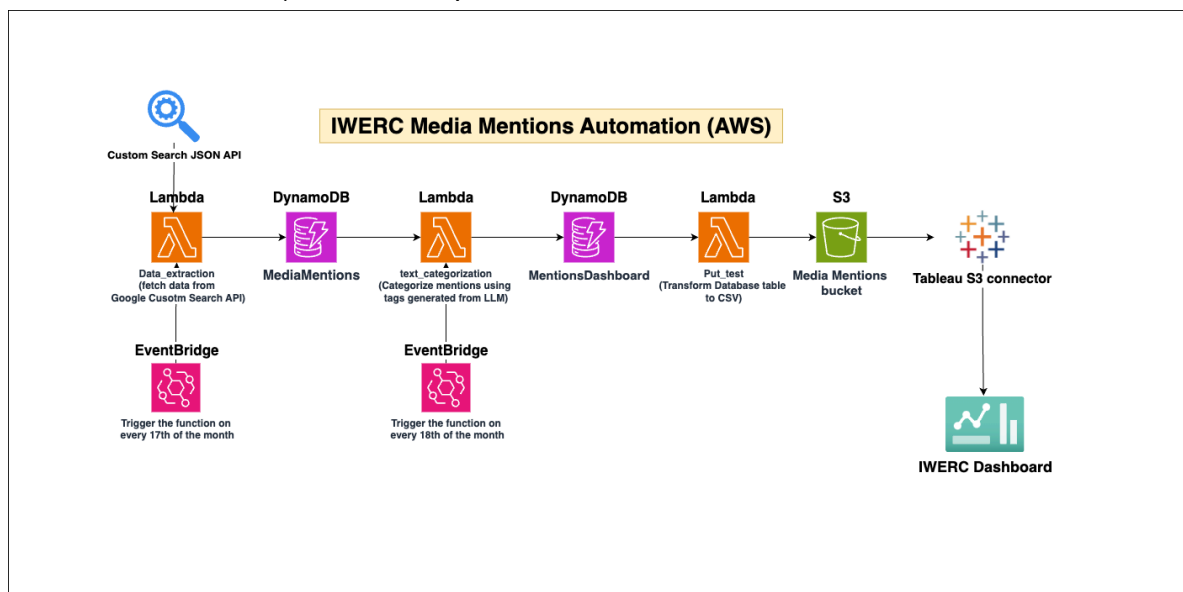
a) The full circle

(1) Need the client to provide the frequency they want the tool to extract data

b) Web form to manually input publication and presentation details



c) Extraction part for media mention & social media



B. Explanation

1. AWS offers many services for us to host and run the pipeline fully on their platform.

How many times it gets viewed or downloaded

Once a quarter

2. All data will be saved in S3
3. Pipeline can then also be integrated with Tableau or Amazon QuickSight.
4. Given our data size isn't large, the cost of using AWS will be minimal.

III. **Solution 1 vs Solution 2**

	Solution 1	Solution 2
Pros	- Lower cost - Low maintenance effort	- Scalable environment - Security - Automation - Real-time data processing
Cons	- Manual effort - Tools only run when there is an execution - Only good with small data sets	- High maintenance effort - Cost overrun if not carefully monitored

IV. **Dashboard details**

A. Tracking dashboard

1. Metrics mentioned
 - a) Publication
 - (1) Publication per educational level
 - (2) Count of publications per portfolio
 - b) Presentation
 - (1) Number of presentations in government sector and public sector
 - (2) Number of presentations for the last three years
 - c) Media mention
 - (1) Number of media mentions per categories
 - (2) Number of media mentions by quarter
 - d) Social media
 - (1) Total impressions by month
 - (2) Total impressions for LinkedIn
 - (3) Total impressions for Blue Sky
2. Highlight features
 - a) Filter by year (show the most recent three years)

B. Website dashboard

1. Metrics mentioned
 - a) Publication

- (1) Publication per educational level
 - (2) Count of publications per portfolio
 - b) Presentation
 - (1) Locations on the map where the presentations were hosted
 - c) Media mention
 - (1) Number of media mentions per categories
 - (2) Number of media mentions by quarter
 - d) Social media
 - (1) Total impressions by month
 - (2) Prediction of impressions for LinkedIn for the next month
 - (3) Prediction of impressions for Blue Sky for the next month
- 2. Highlight features
 - a) Filter by year (show the most current year)
 - b) Highlight the top funders/ partners

C. Need the client's feedback on the metrics to track on the dashboards

Developer account